

MICHAEL LUSTIG

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PROFESSIONAL SUMMARY

Communication-minded Senior Software Engineer specializing in AI-native, agent-addressable systems. I use clean architectural principles to build applications that are easy for both humans and LLM agents to understand and control. By maintaining a clear North Star and making intentional architectural decisions, I substantially accelerate development and debugging by allowing business logic to run independently of heavy frameworks - locally, in CLIs, and in CI. This approach prevents the code duplication and drift that occurs when new functionality is retrofitted onto existing systems, especially LLM-driven functionality that introduces agentic behavior.

EXPERIENCE

Shopify

March 2026 – July 2026

Senior Software Engineer - React Native

Remote

- Joined Shopify's Core Mobile team to help transition mobile experiences toward AI-native architecture while also delivering merchant-facing product work.
- Prototyped AI-native mobile architecture patterns that made app state and actions addressable by LLM agents across login, home, product list, and product detail flows.
- Built local Sidekick developer tooling so a development agent could exercise Shopify in-app AI behavior through the mobile WebView.
- Led mobile delivery for Flex Collections details, working through API versioning, GraphQL payload, header, and backend readiness issues.
- Modularized identity, authentication, product list, and product detail logic for local and CI execution outside the full mobile binary.
- Technologies: React Native, Swift, TypeScript, GraphQL, CLI Tooling

Oscar Health

April 2025 – March 2026

Senior Software Engineer

Remote

- Worked on Oscar member and CareGuide experiences, with a major focus on LLM-assisted workflows, internal search relevance, and Super Agent member interaction systems.
- Built web-to-app download banners that improved app downloads by roughly 230% over the trial period.
- Designed and built Rosco member search, a context-aware CareGuide search experience that improved call resolution by roughly 12% in A/B tests and reduced average call time by roughly 2.5 minutes from an approximately 38-minute baseline.
- Led Follow-Up Actions for Super Agent, Oscar's LLM-driven member interaction system, across Python, gRPC, React, and React Native surfaces.
- Helped set up self-hosted Langfuse for LLM observability, tracing, cost tracking, analytics, and feedback collection.
- Technologies: React, React Native, Python, Flask, Jinja, gRPC, Protobuf, Langfuse, OpenTelemetry

Ephemera

September 2024 – February 2025

Staff Software Engineer - React Native

Remote

- Worked on XMTP and Convos. XMTP is an extensible decentralized message transfer protocol, and Convos was an open-source, extensible, open-protocol messaging application with privacy and encryption as core product constraints.
- Encapsulated XMTP SDK consumption behind a dependency layer and dedicated client management class so protocol implementation details no longer leaked throughout the application.
- Extracted complex wallet creation, identity management, and cross-device linking coordination from tangled React code into XState machines with dependency injection.

- Designed a responsive chat composer with user search, Ethereum address search, group search, optimistic recipient lookup, and inline conversation creation.
- Led Firebase App Check integration across React Native, Android, iOS, and backend systems to increase trust that backend requests came from legitimate application binaries.
- Championed ADRs and used LLM-assisted transcript workflows to turn architecture discussions into agent-readable decision records.
- Technologies: Expo, Privy, thirdweb, Noncustodial Wallets, Smart Contract Wallets, Passkey Authentication, XMTP, Firebase App Check, JWT Authentication, React Native, TypeScript, React, React Native Testing Library, Jest, React Query, React Navigation, zod

Jabra Hearing

August 2023 – September 2024

Senior React Native Engineer

Remote

- Worked on React Native hearing-aid application architecture, Bluetooth pairing reliability, state-machine modeling, state sharing, dependency management, testing, and demo-mode support.
- Systematically diagnosed and resolved Bluetooth pairing issues caused by conflicting parent and child XState machines entering loops.
- Designed and developed a proprietary state management library similar in spirit to early Zustand, with Zod validation, persistence, migrations, and pure JavaScript runtime support.
- Replaced leaky React-dependent static classes with clean observable stores and fakeable dependencies, simplifying tests and enabling demo mode.
- Directed the XState v4 to v5 upgrade across approximately 25 to 30 state machines with zero production regressions.
- Established canonical best practices for XState architecture and championed code architecture and testing practices.
- Technologies: React Native, TypeScript, XState, RxJS, Actor Model, Bluetooth, Jest, React Testing Library, zod, Unleash

Bakkt

August 2020 – March 2021

Fullstack Software Engineer

Remote

- Built React Native and Node.js systems for Bakkt Wallet and application gateway work, with an emphasis on design systems, refactoring, and native wallet integrations.
- Led development of Bakkt Wallet and Application Gateway Proxy.
- Led refactoring efforts and standardized best practices.
- Collaborated with VP of Design for scalable DesignOps.
- Applied finite state machines for complex interactions.
- Implemented native Objective-C and Java modules for React Native to interface with Google Wallet and Apple Pay.
- Technologies: Node.js, React, React Native, Apollo GraphQL, TypeScript, Yarn Workspaces, Figma, Storybook

TCT Ministries

January 2015 – July 2020

Android, iOS, React Native Developer

Remote

- Led cross-platform media catalog and streaming applications across Android, iOS, React Native, and backend integrations.
- Led cross-platform media catalog and streaming applications development.
- Reduced codebase size using React Native and GraphQL.
- Built greenfield software for media applications.
- Managed Google Play Store and Apple App Store listings.
- Migrated legacy Objective-C iOS and Java Android code bases to React Native.
- Interfaced with Senior VP to understand product direction, customer needs, and implementation timelines.
- Technologies: Python, Node.js, iOS Development, Swift, Kotlin, Java, Objective-C, Android Development, React Native, Apollo GraphQL, Brightcove SDK, Android TV, Firebase, Expo

NOTABLE PROJECTS

NoIssue

January 2023 – January 2023

Project management software integration component

- Convenient front-end tool that integrates with GitHub repositories to provide a comprehensive overview of issues from within a web app.
- Link: <https://github.com/blueberrychopsticks/2023xstate-hack-odevs>
- Technologies: XState, React, Github API

Proof of Listening

2023 – 2023

Solana hackathon project

- Cross-platform audio recognition system with Solana NFT minting capabilities and dynamic mint pricing based on on-chain metadata.
- Link: https://youtu.be/OMApXzEV_co?t=58
- Technologies: Solana, Rust, React Native, TypeScript, Firebase, ShazamKit, Swift Kitura, Python, SwiftUI

mtnBooth Twitter Solana Pay Integration

2022 – 2022

Web application

- Web application for minting mtnFrens NFTs integrated with Twitter.
- Technologies: Next.js, Open Graph, Solana, Solana Pay

Breed Up

2018 – 2019

Pet breeding management application

- Location-based dog owner social platform with pet breeding management features.
- Link: <https://github.com/technoplato/BreedUp>
- Technologies: React Native, Firebase, Firebase Storage, Geojson, TypeScript, react-native-photo-picker

Beauty Time

2019 – 2020

Mobile application for aestheticians' timer management

- Mobile application for managing multi-step lash and brow treatment timers.
- Link: <https://github.com/technoplato/BeautyTime>
- Technologies: React Native, Expo, Redux, Redux Saga, TypeScript, Firebase, Git

Cookies By Us

2017 – 2017

E-commerce platform

- Platform connecting custom bakers with clients, featuring private chat and order customization.
- Technologies: React Native, Node.js, Firebase

TV Portal

May 2012 – September 2016

Top-charting Android application with 2M+ downloads

- Top-charting Android app with over 2M downloads, 27,000+ social followers, and nearly 100 GitHub stars.
- Link: <https://github.com/tvportal/android>
- Technologies: Android, Play Store, Third Party App Store Management

GPT Custom Server

2024 – 2024

AI development tool

- Early local AI tooling that enabled ChatGPT-style workflows to interact with a local filesystem and run tests.
- Technologies: TypeScript, Express, Docker, ngrok, Bun, Jest, Bash

Organic Software

2023 – Present

iOS development tool

- SwiftUI and TCA application for real-time speech transcription, LLM processing, code modifications, and PR generation.

- Technologies: SwiftUI, The Composable Architecture, OpenAI APIs, WebSockets, Apple Speech Framework

Android Open Source Benchmark Suite

2015 – 2016

IEEE publication and Android benchmark work

- Modified open source Android applications for a deterministic mobile benchmark suite connected to an IEEE publication.
- Link: <https://ieeexplore.ieee.org/document/7482089>
- Technologies: Android, Java

TECHNICAL SKILLS

AI & Agentic Systems	Agentic Engineering, AI-native Architecture, AI Addressability, LLM Integration, LLM Observability, Local Agent Tooling, Skill Creation, Skill Maintenance, Architectural Decision Records
Product & Architecture	Communication-minded Engineering, North Star Alignment, Clean Architecture, Business Logic Outside UI, CLI-exercisable Logic, Dependency Injection, Dependency Isolation, Modular Architecture, Cross-team Collaboration
Programming Languages & Core Technologies	TypeScript, JavaScript, Python, Swift, Java, Kotlin, Objective-C, Rust, Node.js, Bun, GraphQL, REST, Neo4j, Docker, RxJS, zod
Frontend Development	React, React Native, Next.js, SwiftUI, XState, Redux, Redux Saga, React Query, Jest, React Testing Library, React Native Testing Library, Cypress, Percy, Chromatic, Rollup, Webpack, Storybook, Figma, React Navigation, Apollo GraphQL
Mobile Development	React Native, Expo, Swift, SwiftUI, The Composable Architecture, Android SDK, Google Play Store, Apple App Store, Android TV, Brightcove SDK
Backend & Infrastructure	GraphQL Schema Design, REST API Design, Firebase, Firebase AppCheck, Neo4j, AWS, GitHub Actions, CI/CD, Git, GitHub, JWT, Yarn Workspaces, Unleash, Microservices, Distributed Systems
Web3 & Blockchain	Solana, Ethereum, Solana Pay, Smart Contracts, Anchor, XMTP, Noncustodial Wallets, Smart Contract Wallets, thirdweb, Privy, Passkey, NFTs, Blockchain Integration

EDUCATION

Florida State University

2011 – 2015

B.S. in Computer Science, Computer Science

- Minor in Mathematics and Physics
- Focus: Mentoring/Teaching

Udacity

2014 – 2014

Android Development Nanodegree, Android Development

- Android Development Nanodegree

AWARDS & RECOGNITION

NoIssue - Winner - XState Orlando Devs 2023 Hackathon (January 2023) - Orlando Developers Meetup & statel.ai. Developed a convenient front-end tool that integrates with GitHub repositories to provide a comprehensive overview of issues from within your web app.

Inaugural Android Developer Career Summit Attendee (November 2015) - Google. Selected by Google from thousands of students through Udacity's Android Developer Nanodegree program. Attended exclusive summit in Mountain View, participated in themed hackathon and networking events.